

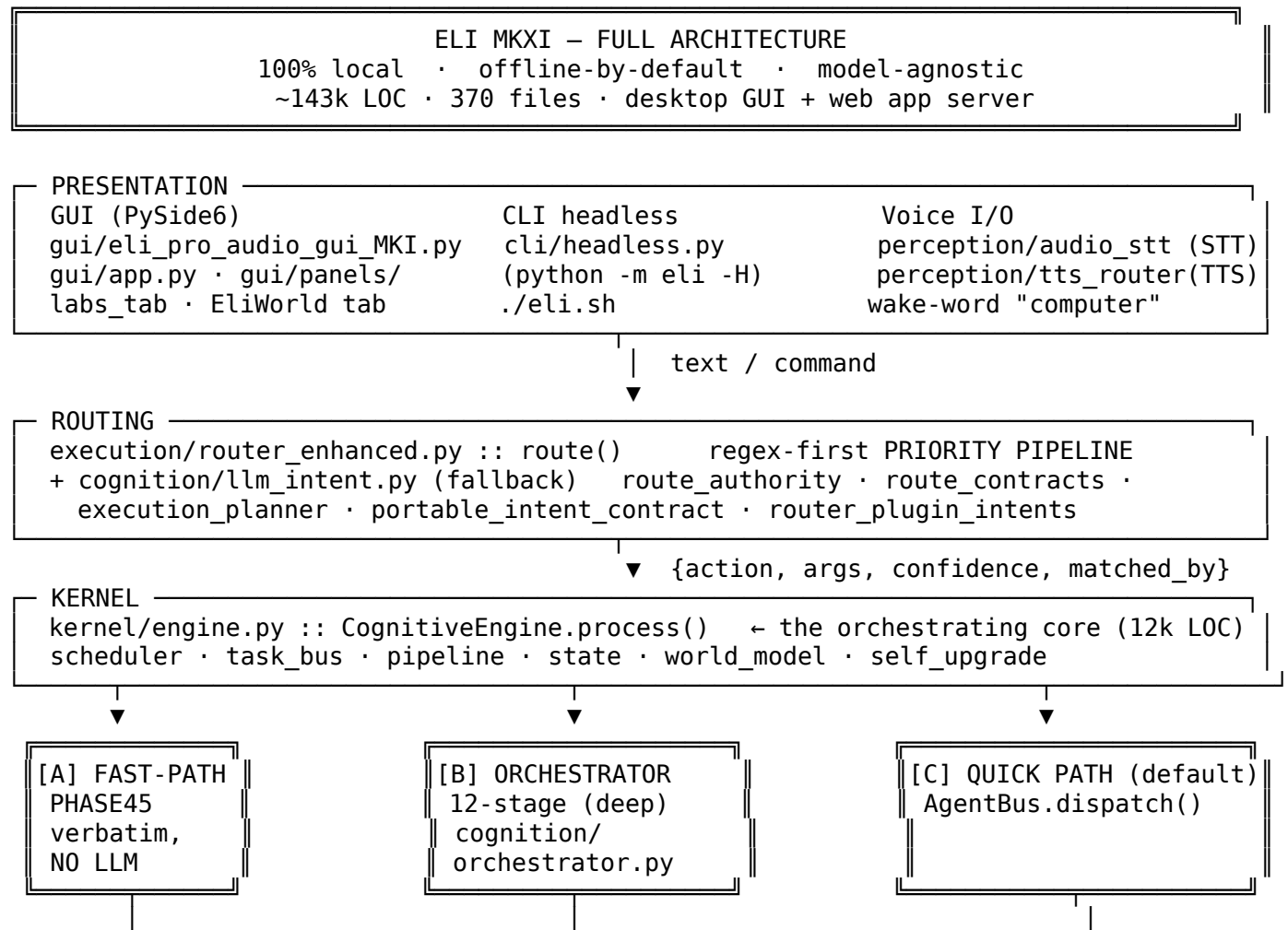
Blueprint — ELI MKXI Full Architecture (ASCII)

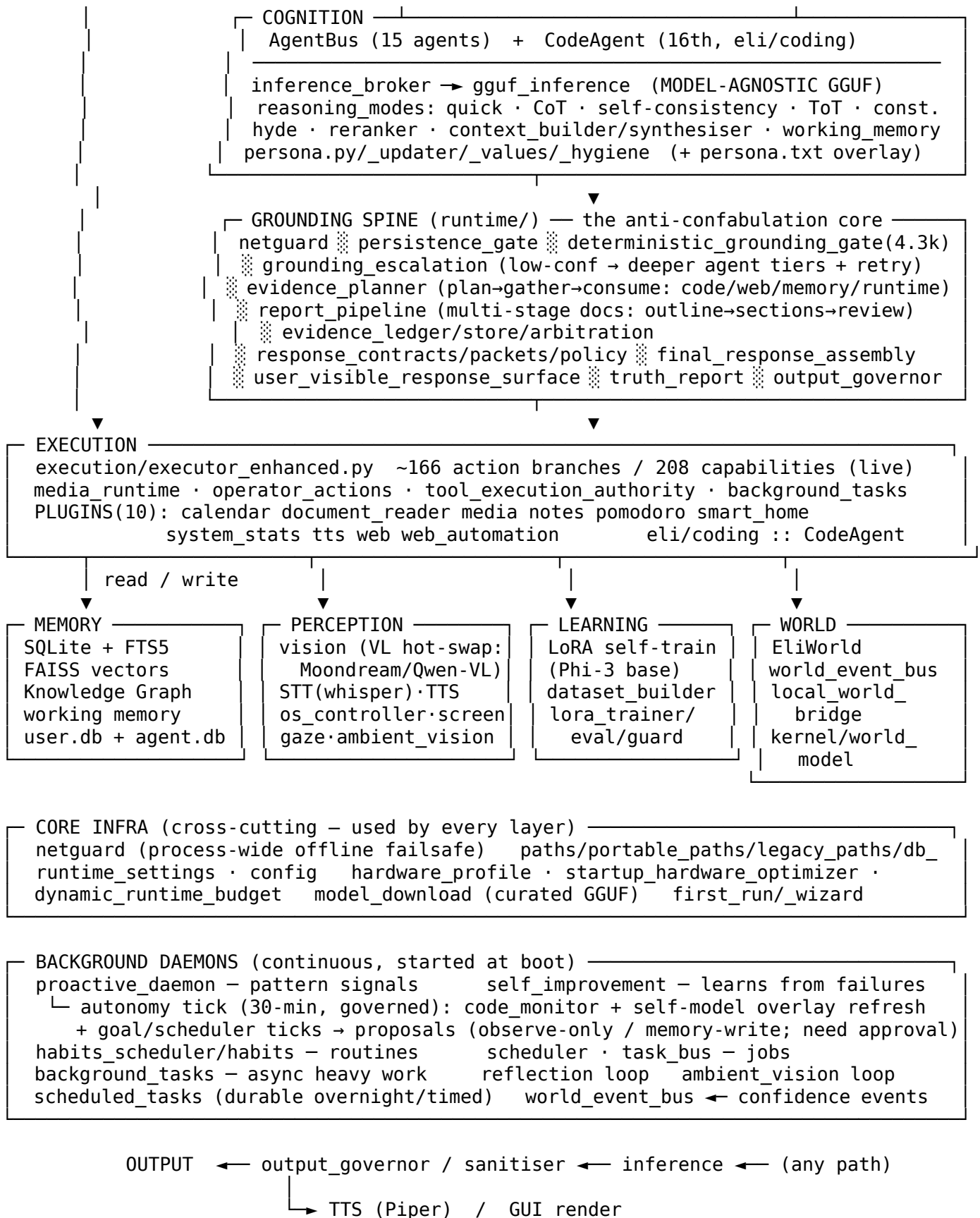
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The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

A. Full system — layered block diagram





B. The 15 agents

AGENT BUS (cognition/agent_bus.py :: AgentBus.dispatch → DispatchResult)
selects a set → runs on a dependency DAG → aggregates grounding_confidence

BusMemoryAgent	recall: SQLite/FTS5/FAISS → context
KnowledgeGraphAgent	entities/relations from the KG
SystemAgent	runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)
OrchestratorAgent	plan / coordinate grounded synthesis
FileCodeAgent	searches the codebase for code-grounded answers
IntrospectionBus...	live runtime + gathers identity/awareness audits as evidence (persona summarises – never a data dump)
CapabilityAgent	capability manifest (what ELI can do)
ReflectionAgent	session reflection / insights
ProactiveAgent	proactive patterns / signals
HabitAgent	learned habits
SelfImprovementAgent	self-improvement state
FrontierAgent	frontier reasoning hooks
PluginAgent	dispatch to registered plugins
VoiceAgent	STT/TTS engine state
CriticAgent	second-tier verifier: checks retriever agents' output
+ CodeAgent (16th) eli/coding/agent	SEPARATE pipeline: plan→search→verify→repair (CODE_SOLVE / GENERATE_SCRIPT) – not a bus agent

C. Module tree (LOC · key files · role)

eli/ (~143k LOC, 370 files) · api/server.py (FastAPI web app + dashboard)

__main__.py		entry dispatch (GUI --headless)
kernel/	13.1k	— the core
engine.py	12.0k	CognitiveEngine.process() – the spine *god-file
scheduler.py		timed jobs
task_bus.py · pipeline.py · state.py · world_model.py · self_upgrade.py		
execution/	20.5k	— route → act
executor_enhanced.py	14.3k	~174 actions *god-file
router_enhanced.py	6.1k	priority pipeline *god-file
execution_planner.py · route_authority.py · route_contracts.py		
tool_execution_authority.py · operator_actions.py · operator_policy.py		
media_runtime.py · router_plugin_intents.py · executor_plugin_handlers.py		
portable_intent_contract.py · execution_intent_packets.py		
cognition/	12.3k	— think
agent_bus.py	2.4k	15 agents + dispatch *
orchestrator.py		12-stage deep retrieval
gguf_inference.py	2.1k	· inference_broker.py model-agnostic inference
reasoning_modes.py · hyde.py · reranker.py · llm_intent.py		
persona.py/_updater/_values/_status/_hygiene		(+ persona.txt, persona.auto.txt)
context_builder.py · context_synthesiser.py · working_memory.py		
output_governor.py · response_governance.py · response_sanitizer.py		
grounded_status.py · introspection_agent.py · tone_analyzer.py · chat_model.py		
runtime/	18.0k	— grounding spine + daemons (68 files)

- deterministic_grounding_gate.py 4.3k ★ · grounding_escalation.py
- evidence_ledger/store/arbitration.py · memory_evidence.py
- persistence_gate.py · truth_report.py · control_contracts.py
- response_contracts/packets/policy.py · final_response_assembly/provider.py
- user_visible_response_surface.py · personal_memory_*.py · reflection.py
- background_tasks.py · self_improvement.py · code_monitor.py
- capability_sync.py · pending_proposal.py · runtime_policy.py
- memory/ 6.6k — remember (13 files)
 - memory.py 4.5k Memory · SQLite + FTS5 ★god-file
 - vector_store.py FAISS index
- perception/ 5.4k — sense (18 files)
 - vision.py · analyze_image/csv/pdfs/mesh.py · ambient_vision.py
 - audio_stt.py · local_whisper_stt.py · voice_worker(_streaming).py · eli_listen.py
 - tts_router.py · os_controller.py · screen_locator.py · gaze_engine.py
- planning/ 3.1k — proactivity (21 files)
 - proactive_daemon.py · habits_scheduler.py · habits.py · jobqueue_cli.py
- coding/ 1.5k — CodeAgent (plan→search→verify→repair, 9 files)
- learning/ 3.1k — LoRA self-training (11 files)
 - lora_trainer/eval/guard.py · dataset_builder/filters.py
 - bootstrap_phi3_base.py · base_model_resolver.py · training_preflight.py
- plugins/ 1.9k — manager + 10 plugins
 - calendar · document_reader · media · notes · pomodoro · smart_home · system_stats · tts · web · web_automation
- world/ 1.5k — EliWorld (world_event_bus, local_world_bridge)
- core/ 4.9k — infra
 - netguard.py offline failsafe + allow_network()
 - paths.py · portable_paths.py · legacy_paths.py · db_paths.py
 - runtime_settings.py · config.py · grounding.py
 - hardware_profile.py · startup_hardware_optimizer.py · dynamic_runtime_budget.py
 - model_download.py · first_run.py · first_run_wizard.py
- gui/ 18.7k — PySide6 desktop
 - eli_pro_audio_gui_MKI.py 11.0k ★god-file · app.py · labs_tab.py 5.7k
 - panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)
- tools/ 8.1k — image_engine · news · document tools
- contracts/ 0.7k · cli/ 0.1k · system/ 0.3k · utils/ 0.9k

D. On-disk data layout

- artifacts/
 - db/
 - user.sqlite3
 - memories(+_fts) · conversation_turns · conversations · kg_entities(+_fts) · kg_relations · recall_log · runtime_events · news_articles(+_fts) · news_reflections · habits/habit_events/habit_rules · observations · learning_replay · working_memory_pins · user_patterns · session_summaries · corrections · failures

```

├── agent.sqlite3      agent_dispatches · agent_metrics · improvements ·
                        failures · code_patches · error_tracking · observations
├── vectors/index.faiss semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
├── conversations/ + archive/
├── runtime/users/<uuid>/user_profile... per-user profile
├── runtime_snapshot.json live model/runtime truth
├── world/{snapshots,ledger,journal}/ · proactive/ · incidents/<date>.jsonl
├── documents/ · scripts/ · analyze_image_*/
├── config/ settings.json (gitignored) · templates/settings.template.json ·
            settings.example.json · plugins_state.json
├── models/ <your>.gguf · embeddings/ · whisper/ · image/ voices/ *.onnx (Piper)

```

E. The flow in one line

```

INPUT → ROUTER → ENGINE ┌── FAST-PATH ───────────────────────────→ OUTPUT (verbatim)
                        │── ORCHESTRATOR → retrieval ┌──┘
                        │── QUICK → AGENT BUS ───────────┘
MEMORY (SQL/FTS5/FAISS/KG/WM) → context ←──────────┘
GATES: netguard · persistence · grounding · escalation · confidence · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / mem-
ory / gating close-ups) · eval_harness.md (the measurement layer). ""

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